

Novel constraints on the primordial power spectrum from compact object formation

Updated constraints on the primordial power spectrum at sub-Mpc scales

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(Dated: June 25th, 2025)

The primordial power spectrum of matter density perturbations contains highly valuable information about new fundamental physics, in particular cosmological inflation, but is only very weakly constrained observationally for small cosmological scales $k \gtrsim 3 \text{ Mpc}^{-1}$. We derive novel constraints, $\mathcal{P}_\mathcal{R}(k) \lesssim 5 \cdot 10^{-6}$ over a large range of such scales, from the formation of ultracompact minihalos in the early universe. Unlike most existing constraints of this type, our results do not rest on the assumption that dark matter can annihilate into ordinary matter.

I. INTRODUCTION

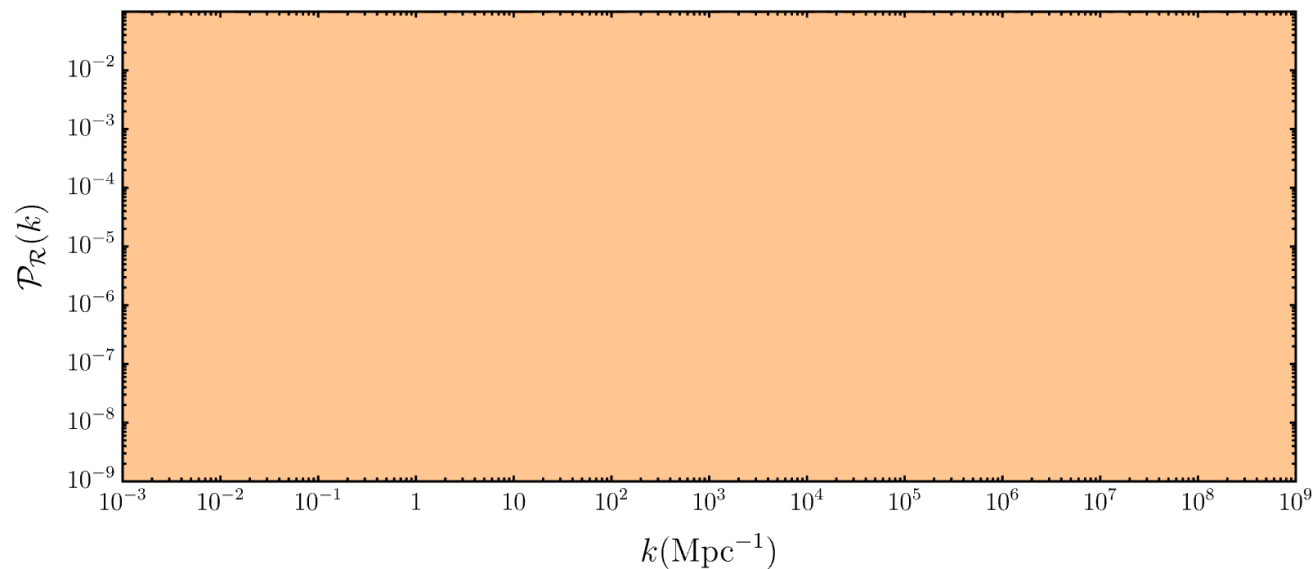
The power spectrum of primordial density fluctuations, $\mathcal{P}(k)$, is a key quantity in cosmology. It describes the initial conditions of the Universe right after the big bang, tiny perturbations in an almost homoge-

background [29, 30] provide complementary constraints directly on the linear, or only mildly non-linear, power spectrum at scales down to roughly one pc.

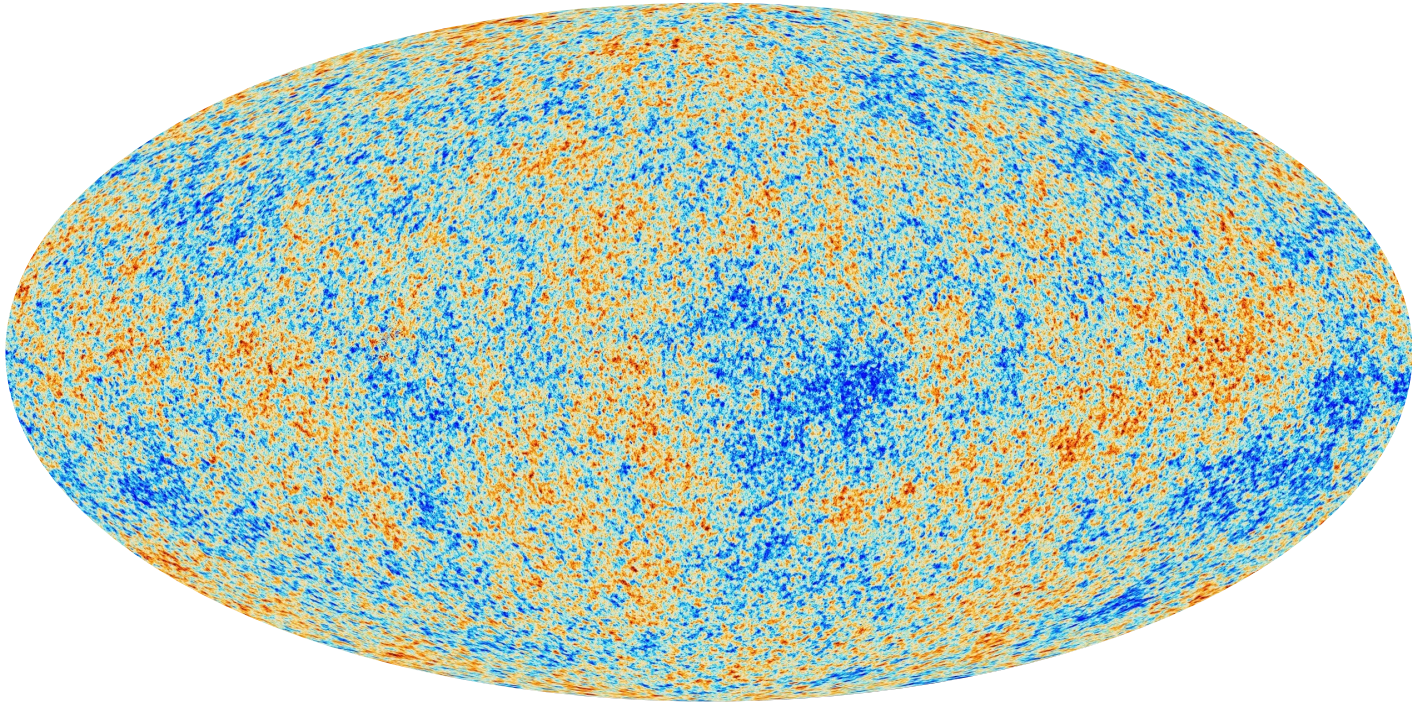
UCMHs are only one example of extended DM objects (EDOs) that may have existed since primordial times. Recently, in fact, the possibility of more general

Our plan for today

Update constraints from the primordial curvature power spectrum

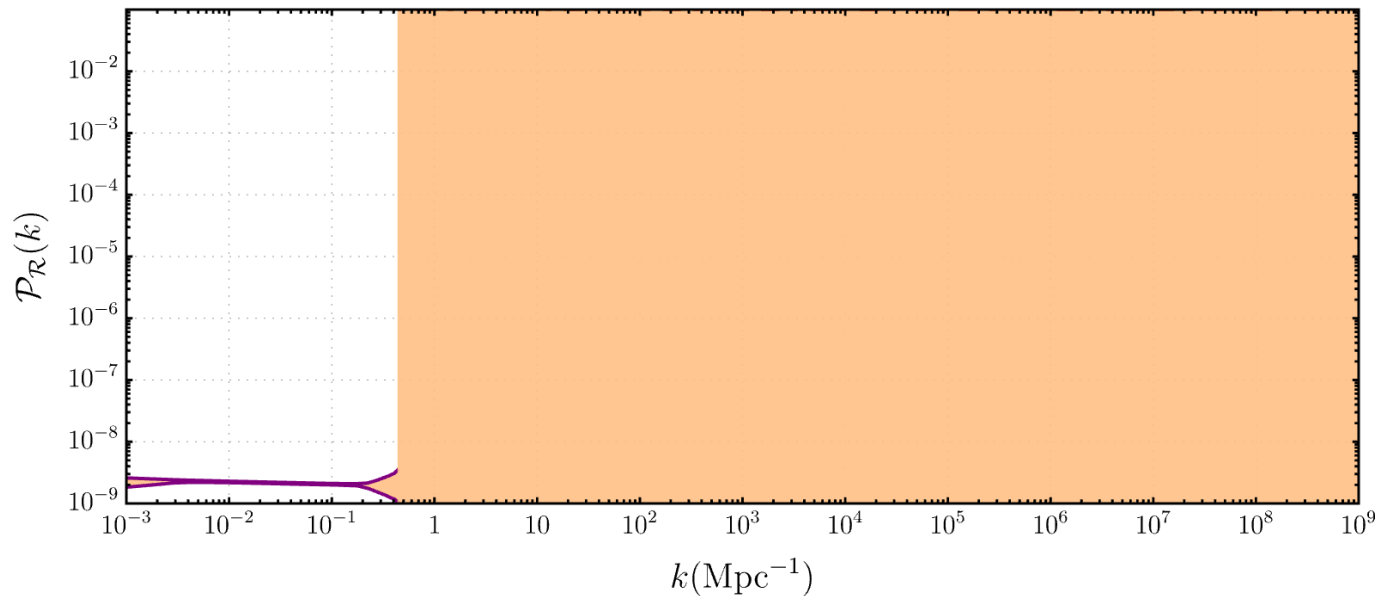


Cosmic Microwave Background



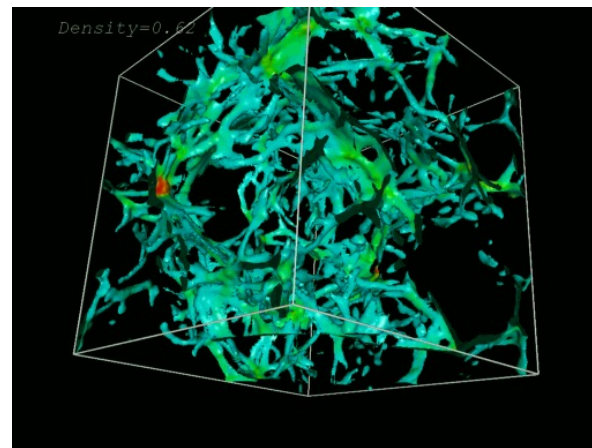
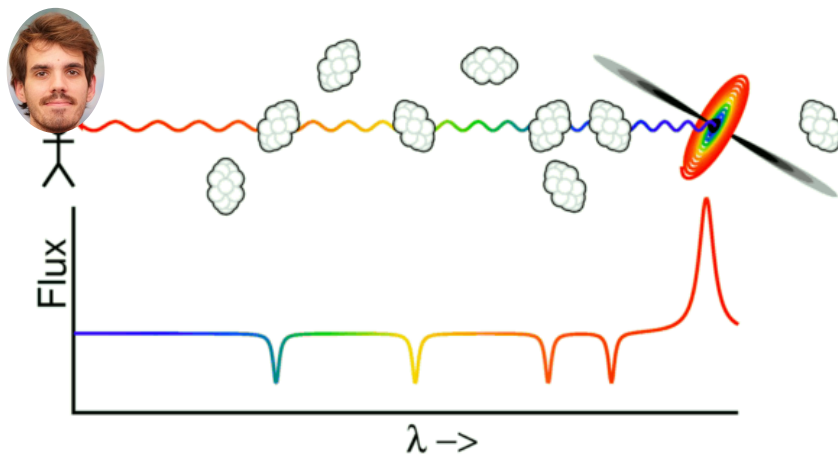
Planck!

Inferring smaller scales by looking at matter distribution in the late universe



Lyman α Forest

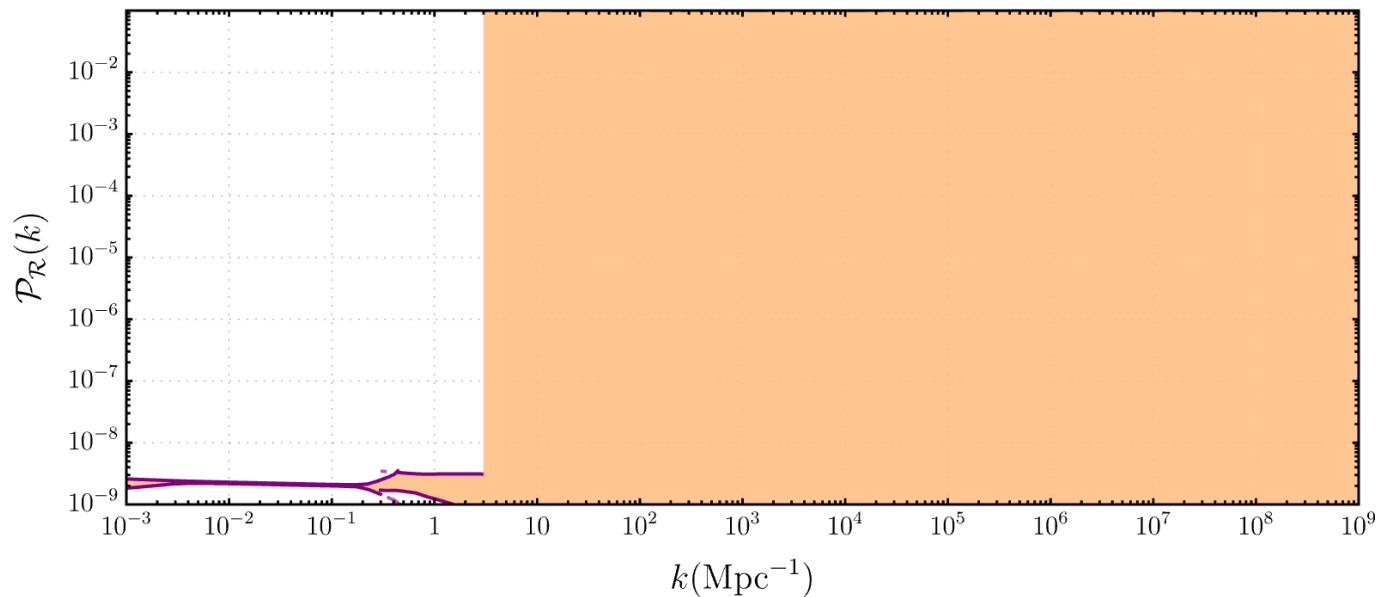
Inferring smaller scales by looking at matter distribution in the late universe



S. Bird, H. V. Peiris, M. Viel, L. Verde (1010.1519v2)

Lyman α and Planck

Inferring smaller scales by looking at matter distribution in the late universe



y and μ distortions

Primordial perturbations create injections of energy into the background that lead to a departure of the perfect black body spectrum:

$$\mu \approx 2.2 \int_{k_{\min}}^{\infty} \mathcal{P}_{\mathcal{R}}(k) \left[\exp\left(-\frac{\hat{k}}{5400}\right) - \exp\left(-\left[\frac{\hat{k}}{31.6}\right]^2\right) \right] d \ln k \quad \mu < 9 \times 10^{-5}$$

$$y \approx 0.4 \int_{k_{\min}}^{\infty} \mathcal{P}_{\mathcal{R}}(k) \exp\left(-\left[\frac{\hat{k}}{31.6}\right]^2\right) d \ln k, \quad y < 1.5 \times 10^{-5}$$

COBE (g605054)

Where we assume a locally scale-invariant power spectrum

$$\mathcal{P}_{\mathcal{R}}(k) = \begin{cases} \mathcal{P}_{\mathcal{R}}(k_{\mathcal{R}}) & \text{for } 1/3 < k/k_{\mathcal{R}} < 3 \\ 0 & \text{otherwise} \end{cases}$$

J. Chluba, A. L. Erickcek, I. Ben-Dayan (1203.2681)

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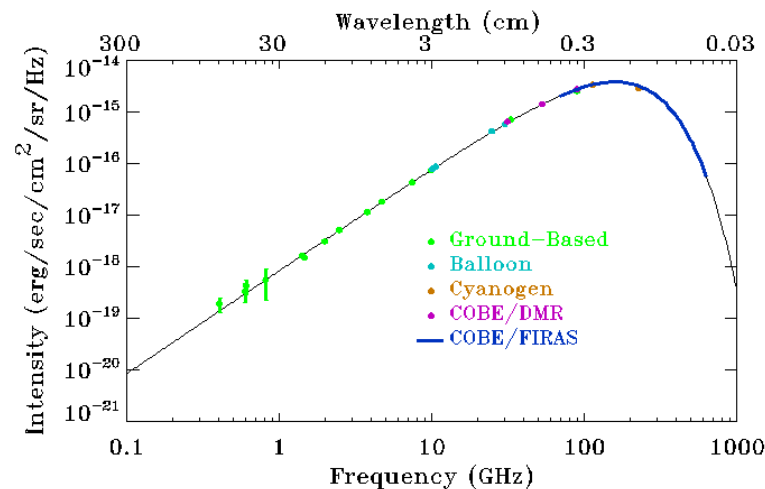
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COBE (9605054)

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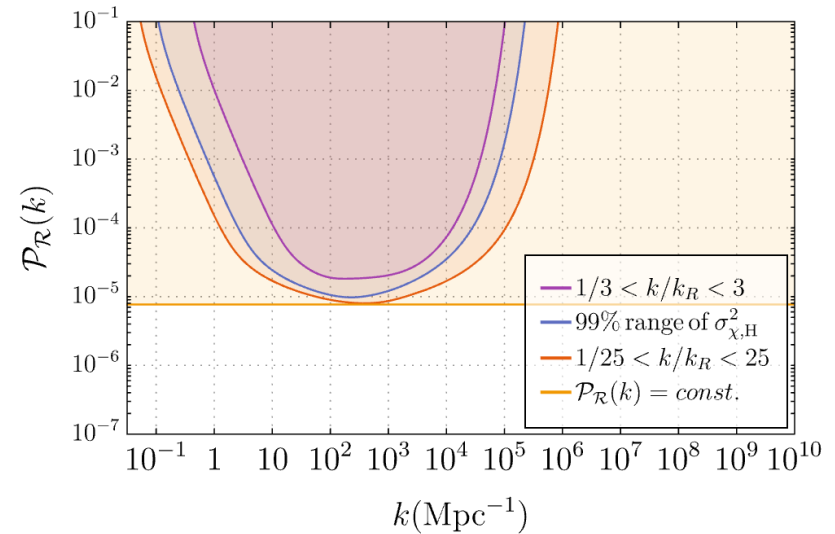
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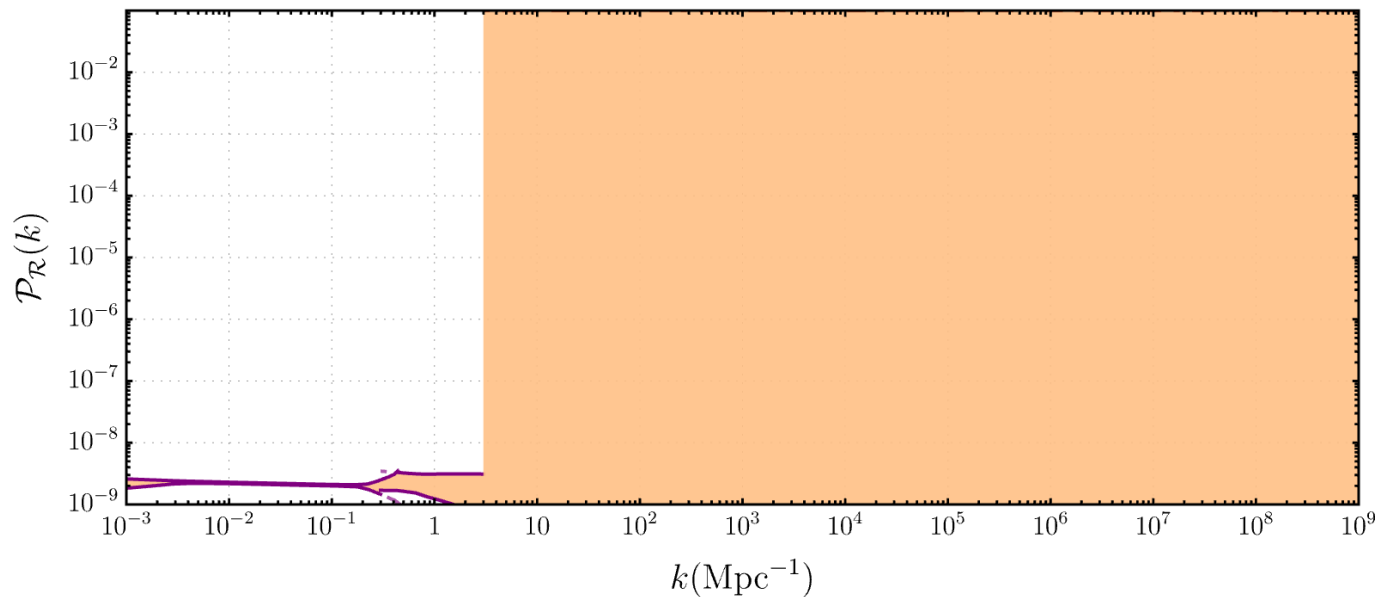


J. Chluba, A. L. Erickcek, I. Ben-Dayan (1203.2681)

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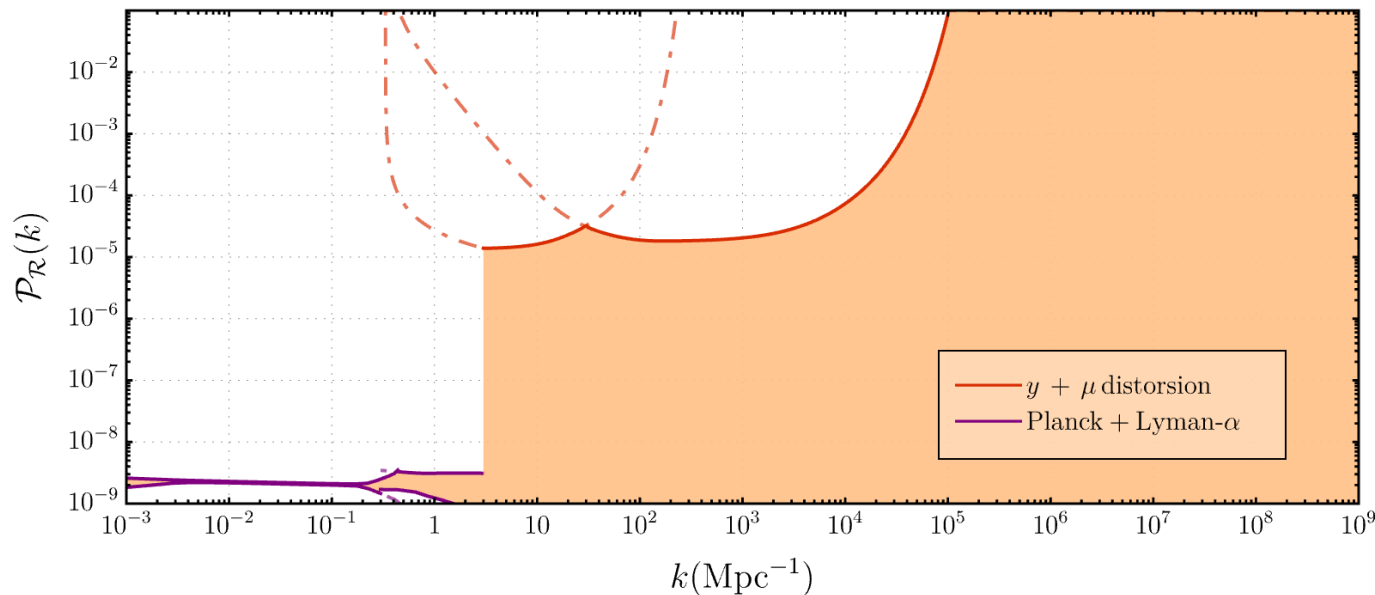
COBE (9605054) J. Chluba, A. L. Erickcek, I. Ben-Dayan (1203.2681)



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COBE (9605054) J. Chluba, A. L. Erickcek, I. Ben-Dayan (1203.2681)

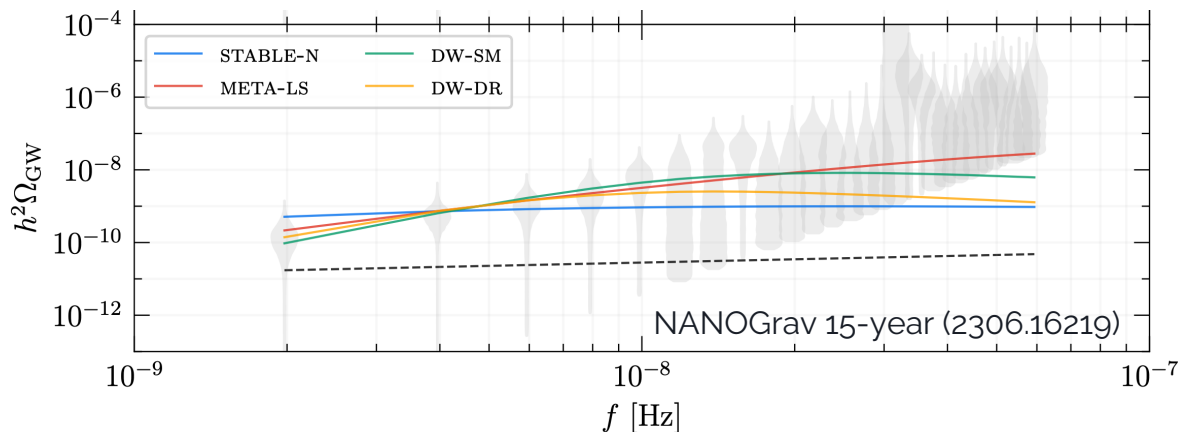


Pulsar Timing Arrays (PTAs)

Any stochastic background of gravitational waves can be constrained via $h^2\Omega_{\text{GW}}$

$$\Omega_{\text{GW}}(k, \eta) = \frac{1}{24} \left(\frac{k}{aH} \right)^2 \mathcal{P}_h(k, \eta) \quad \mathcal{P}_{\mathcal{R}}(k) = \begin{cases} \mathcal{P}_{\mathcal{R}}(k_{\mathcal{R}}) & \text{for } 1/3 < k/k_{\mathcal{R}} < 3 \\ 0 & \text{otherwise} \end{cases}$$

$$\mathcal{P}_h(k, \eta) = 4 \int_0^\infty dv \int_{|1-v|}^{1+v} du \left(\frac{4v^2 - (1 + v^2 - u^2)^2}{4vu} \right)^2 \mathcal{P}_{\mathcal{R}}(ku) \mathcal{P}_{\mathcal{R}}(kv) I^2(u, v, k\eta),$$



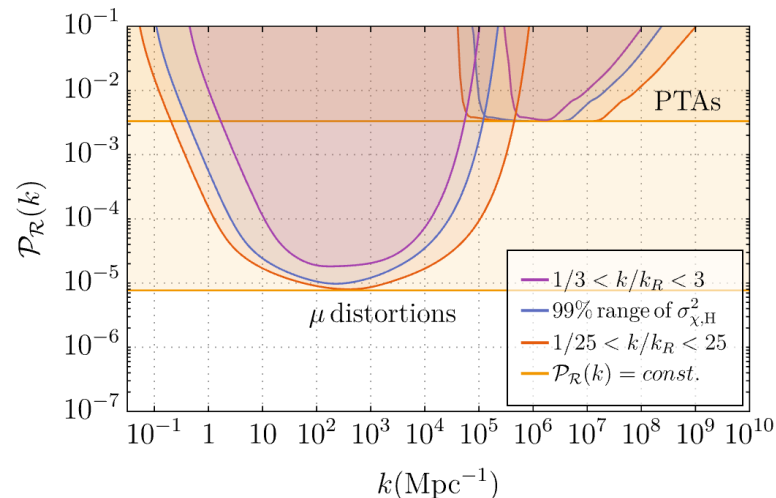
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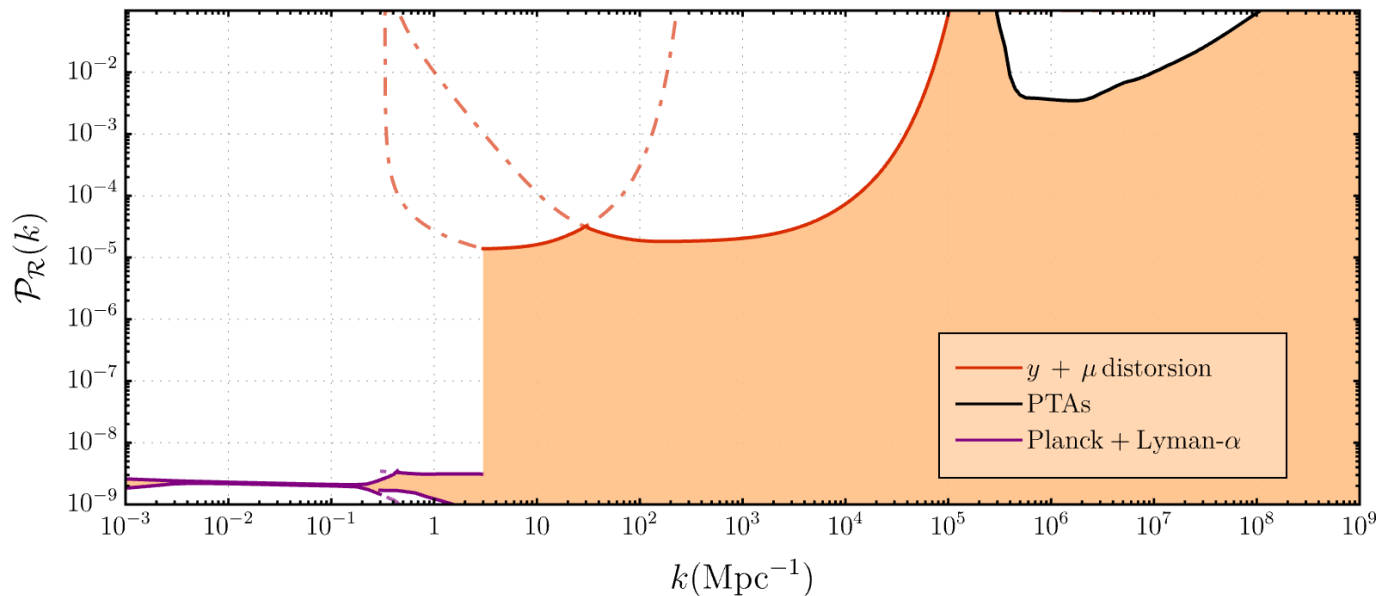


Pulsar Timing Arrays (PTAs)

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NANOGrav 15-year (2306.16219)

$$\Omega_{\text{GW}}(k, \eta) = \frac{1}{24} \left(\frac{k}{aH} \right)^2 \mathcal{P}_h(k, \eta)$$



Primordial Black Holes (PBHs)

Large enough overdensities can lead to the collapse of a region into a black hole!

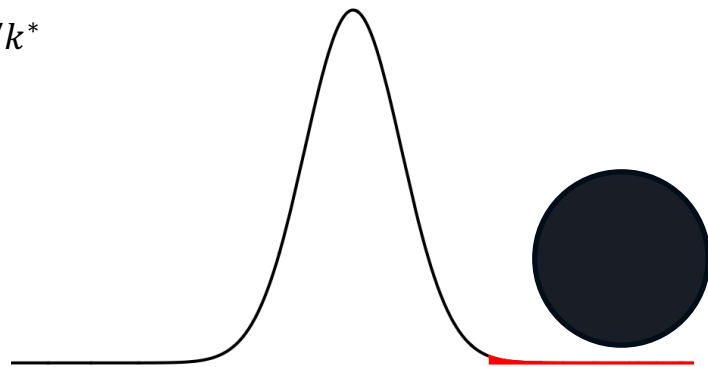
$$\beta(M_{\text{H}}) = \sqrt{\frac{2}{\pi}} \frac{\alpha}{\sigma(R)} \int_{\delta_c}^{\infty} \exp\left(-\frac{\delta^2(R)}{2\sigma^2(R)}\right) d\delta(R)$$

$R = 1/k^*$

$$\delta \equiv (\rho - \bar{\rho})/\bar{\rho}$$

$$\sigma^2(R) = \frac{16}{81} \int_0^{\infty} (kR)^4 W^2(kR) \mathcal{P}_{\mathcal{R}}(k) T^2(kR/\sqrt{3}) \frac{dk}{k}$$

Fraction of black holes

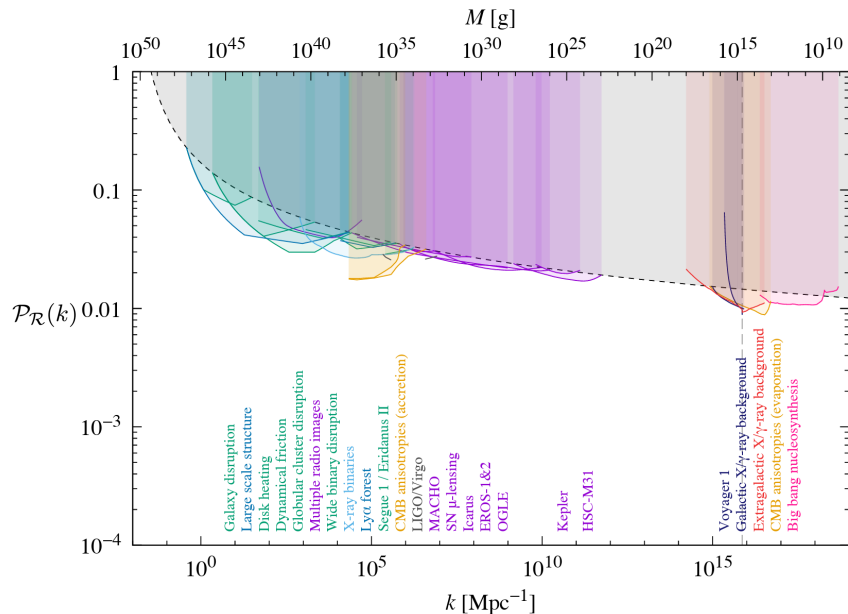
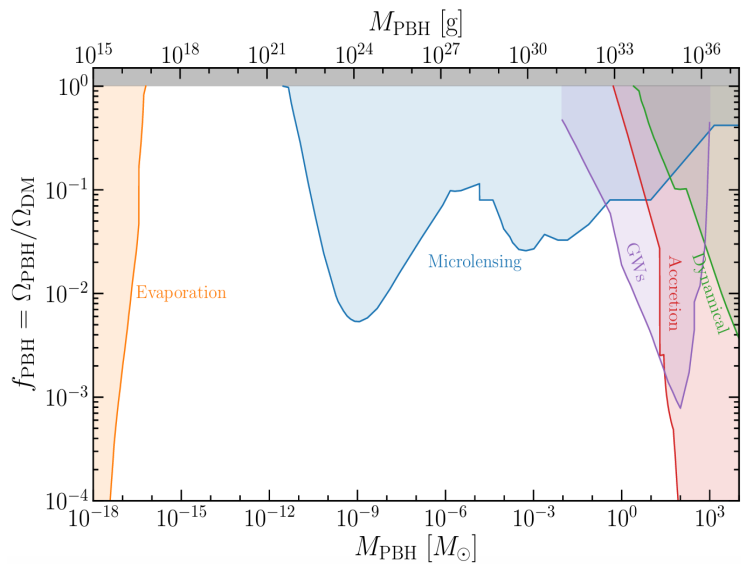


We can therefore constrain the power spectrum using limits on PBHs!

Primordial Black Holes (PBHs)

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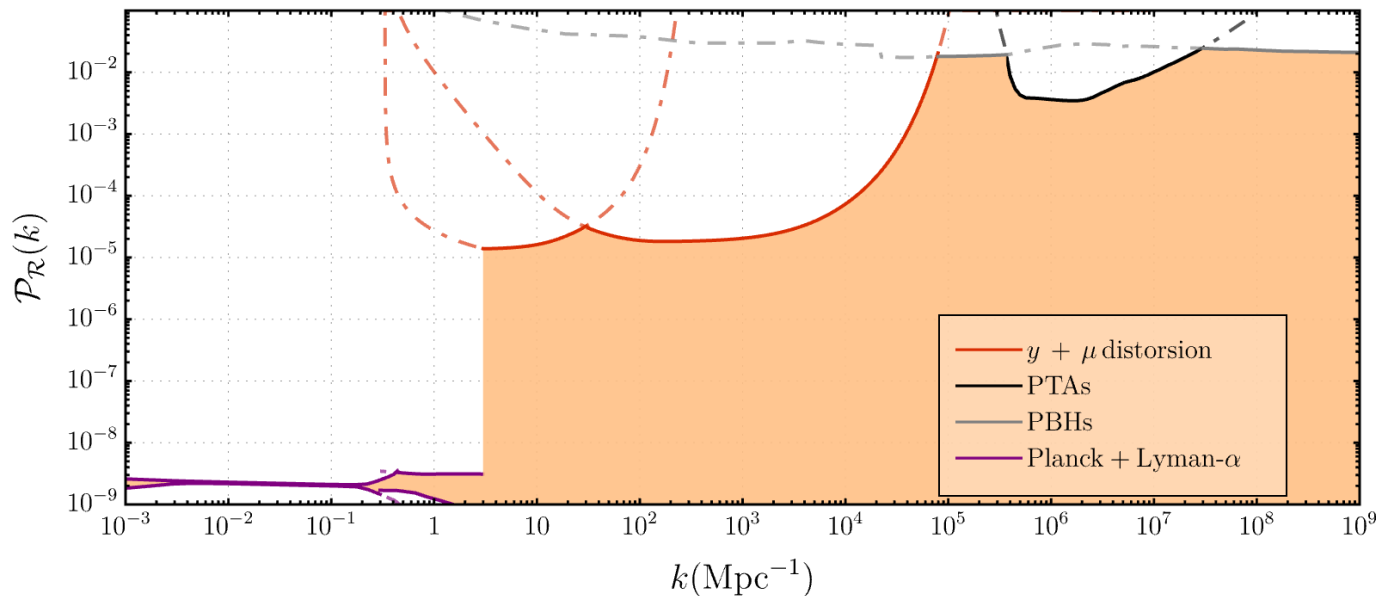
Carr, Kohri, Sndouda and Yokoyama (2002.12778)



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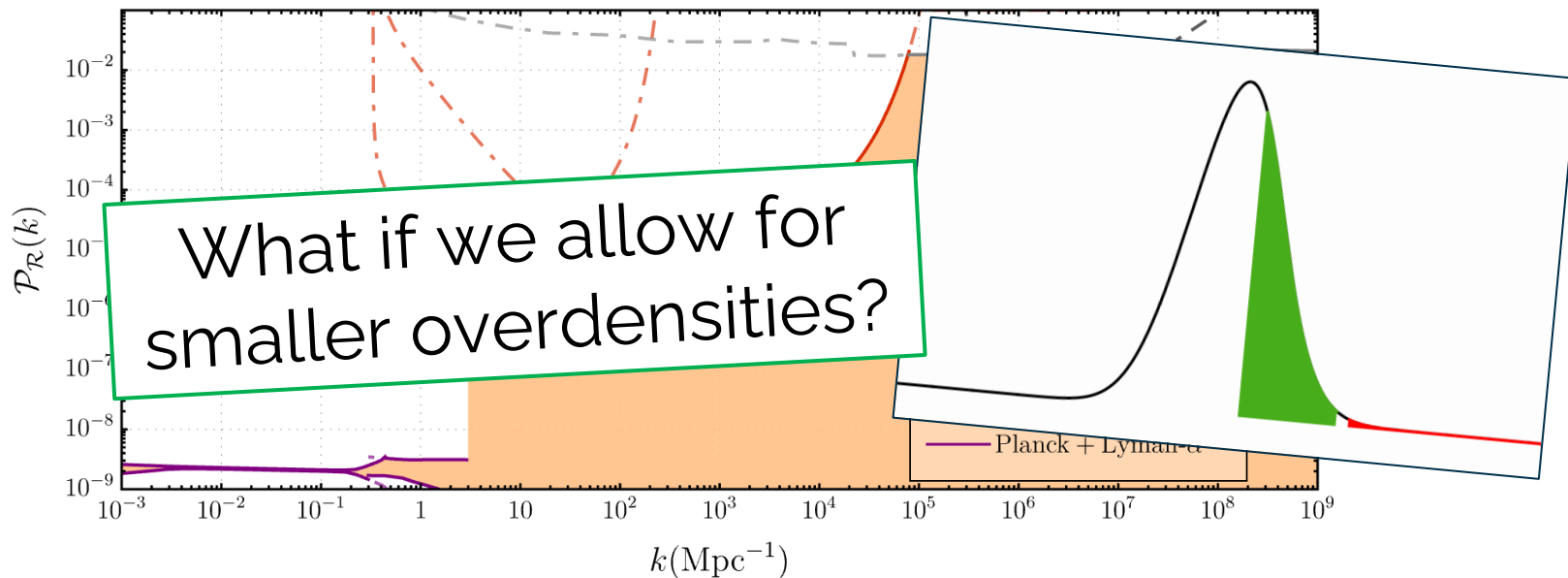
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Ultra-compact minihaloes (UCMHs)

That was the idea from this paper, assuming SM-DM interactions

**Improved constraints on the primordial power spectrum at small scales from
ultracompact minihalos**

Torsten Bringmann*

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Pat Scott[†]

Department of Physics, McGill University, 3600 rue University, Montréal, QC, H3A 2T8, Canada

Yashar Akrami[‡]

*The Oskar Klein Centre for Cosmoparticle Physics,
Department of Physics, Stockholm University,
AlbaNova, SE-106 91 Stockholm, Sweden*

Can we do this only using gravitational interactions?

Ultra-compact minihaloes (UCMHs)

How many are created?

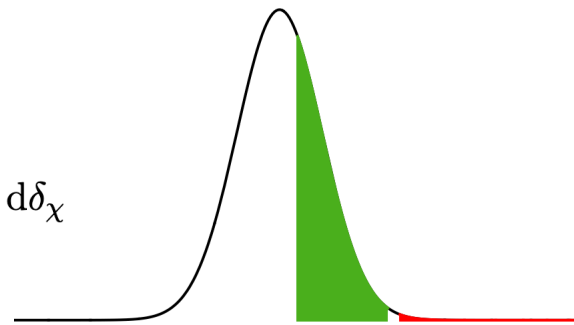
Similar to primordial black holes

$$\beta(k) = \frac{1}{\sqrt{2\pi}\sigma_{\chi,H}(k)} \int_{\delta_{\chi}^{\min}(k)}^{\delta_{\chi}^{\max}} \exp\left[-\frac{\delta_{\chi}^2}{2\sigma_{\chi,H}^2(k)}\right] d\delta_{\chi}$$

$$\Omega_{\text{UCMH}}(k) = \Omega_{\text{DM}}\beta(k)$$

$$\sigma_{\chi,H}^2(k) = \frac{1}{9} \int_0^{\infty} dx x^3 W_{\text{TH}}^2(x) \mathcal{P}_{\mathcal{R}}(xk) T_{\chi}^2(\theta = x/\sqrt{3})$$

$$\delta \equiv (\rho - \bar{\rho})/\bar{\rho}$$



Integrating for smaller overdensities

Ultra-compact minihaloes (UCMHs)

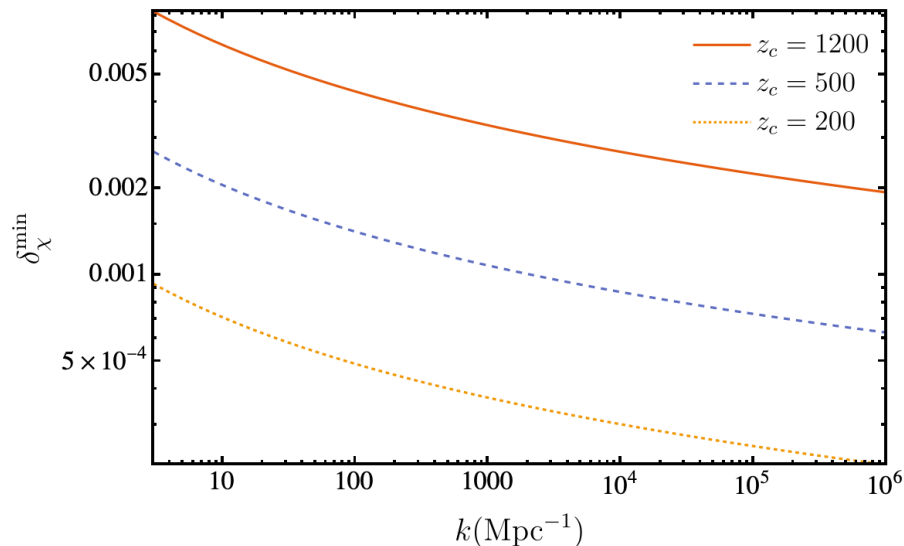
Depending on the scale, different number of objects will be created

$$\Omega_{\text{UCMH}}(k) = \Omega_{\text{DM}}\beta(k)$$

$$\beta(k) = \frac{1}{\sqrt{2\pi}\sigma_{\chi,H}(k)} \int_{\delta_{\chi}^{\min}(k)}^{\delta_{\chi}^{\max}} \exp\left[-\frac{\delta_{\chi}^2}{2\sigma_{\chi,H}^2(k)}\right] d\delta_{\chi}$$

$$\delta \equiv (\rho - \bar{\rho})/\bar{\rho}$$

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Ultra-compact minihaloes (UCMHs)

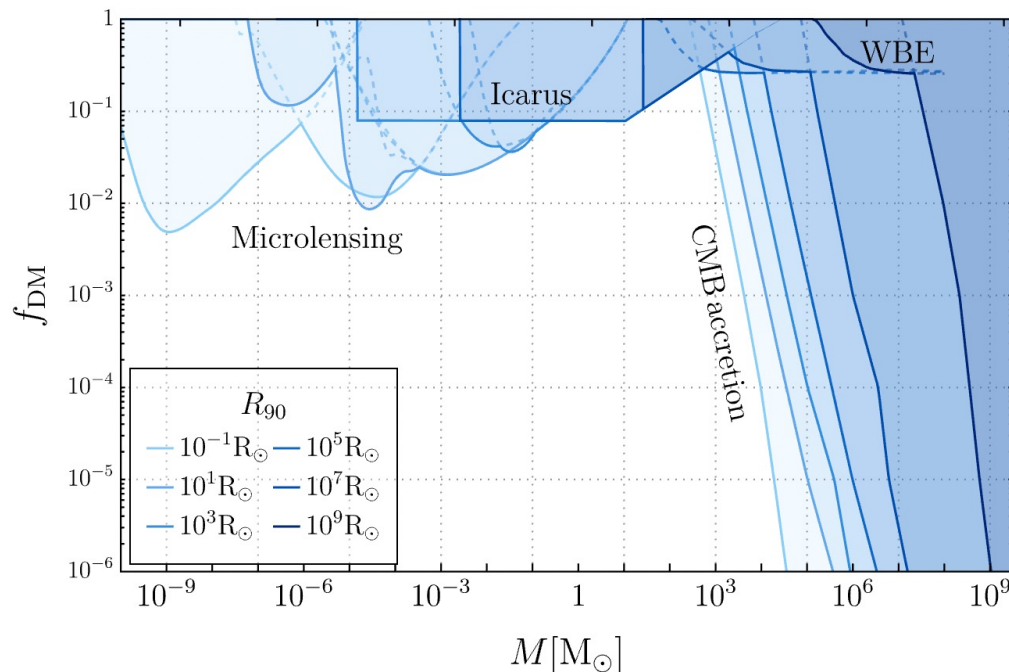
EDObounds:

Choose:

- Shape of the object
- Radius
- Bounds to plot
- Mass distributions!!

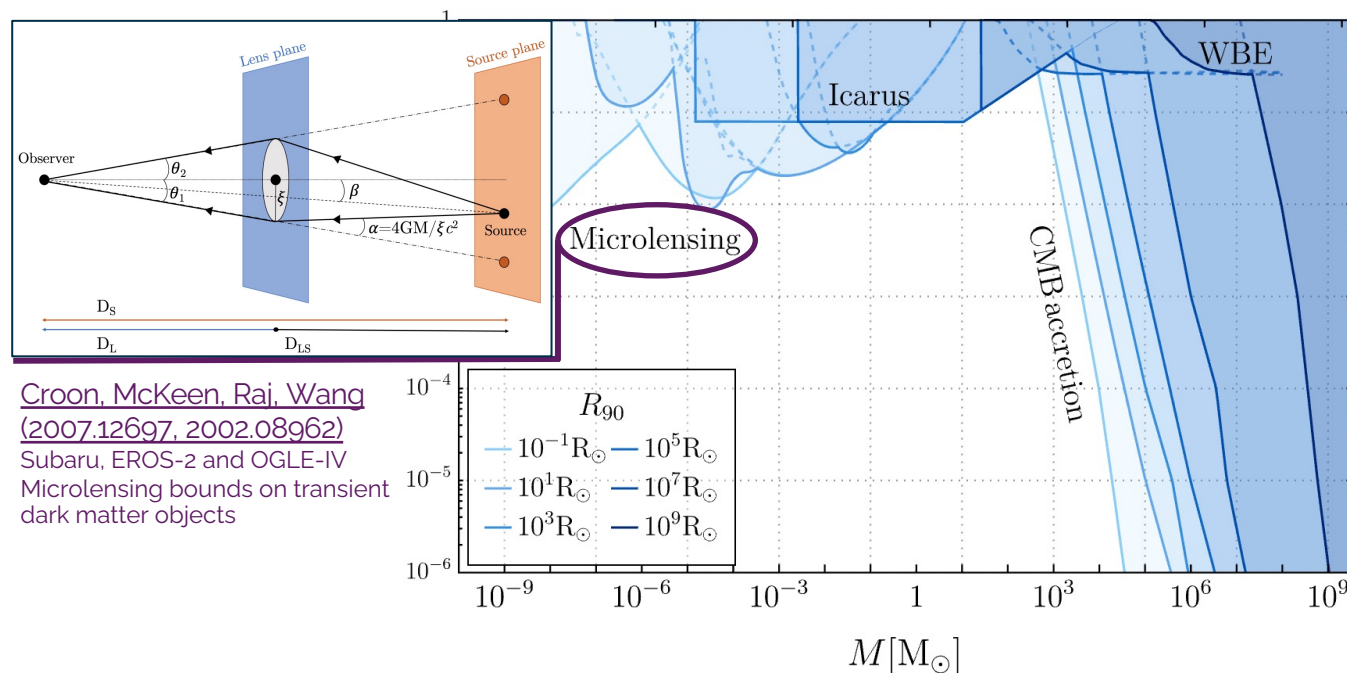
[arXiv:2407.02573](https://arxiv.org/abs/2407.02573)

There has been recent interest on constraining these type of objects:



Ultra-compact minihaloes (UCMHs)

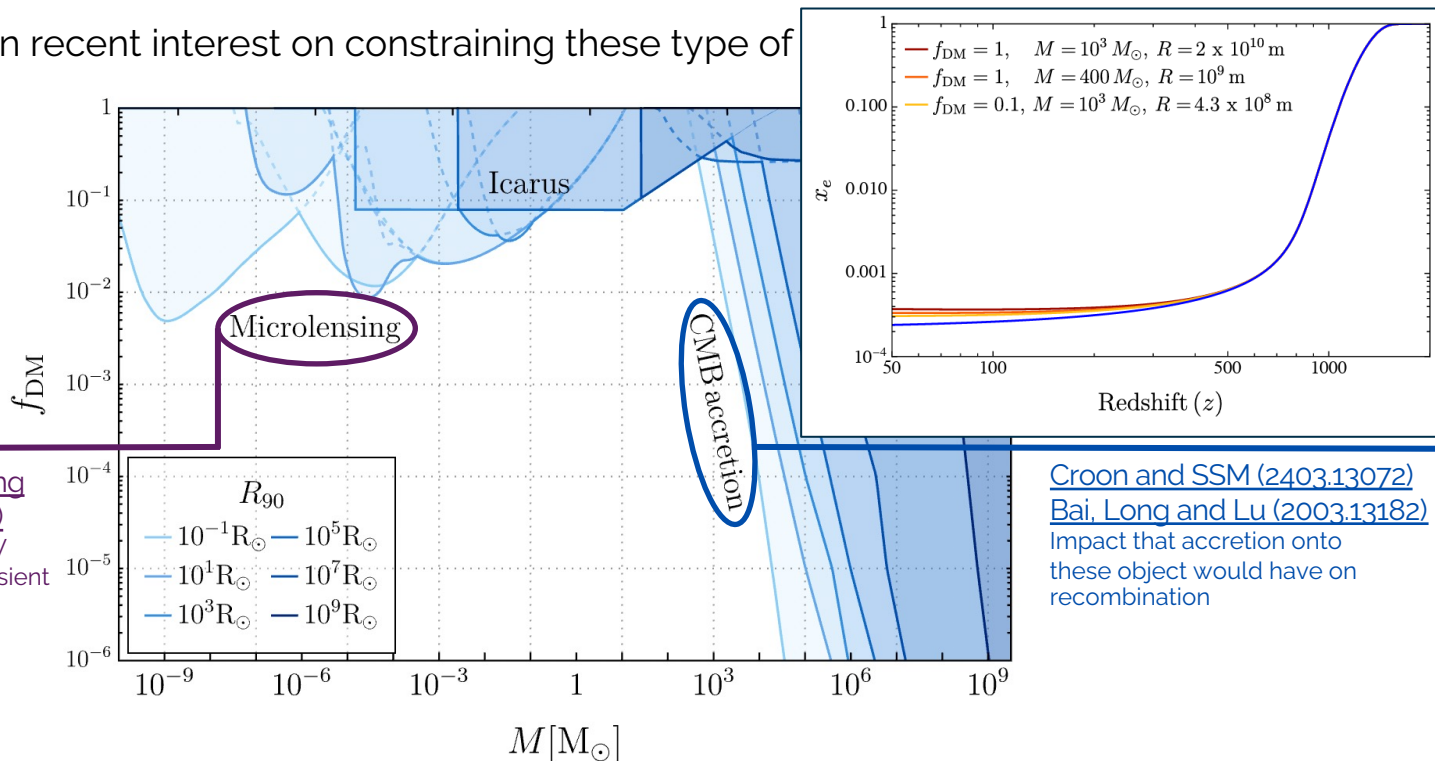
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Ultra-compact minihaloes (UCMHs)

EDObounds:
arXiv:2407.02573

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[Croon, McKeen, Raj, Wang \(2007.12697, 2002.08962\)](#)

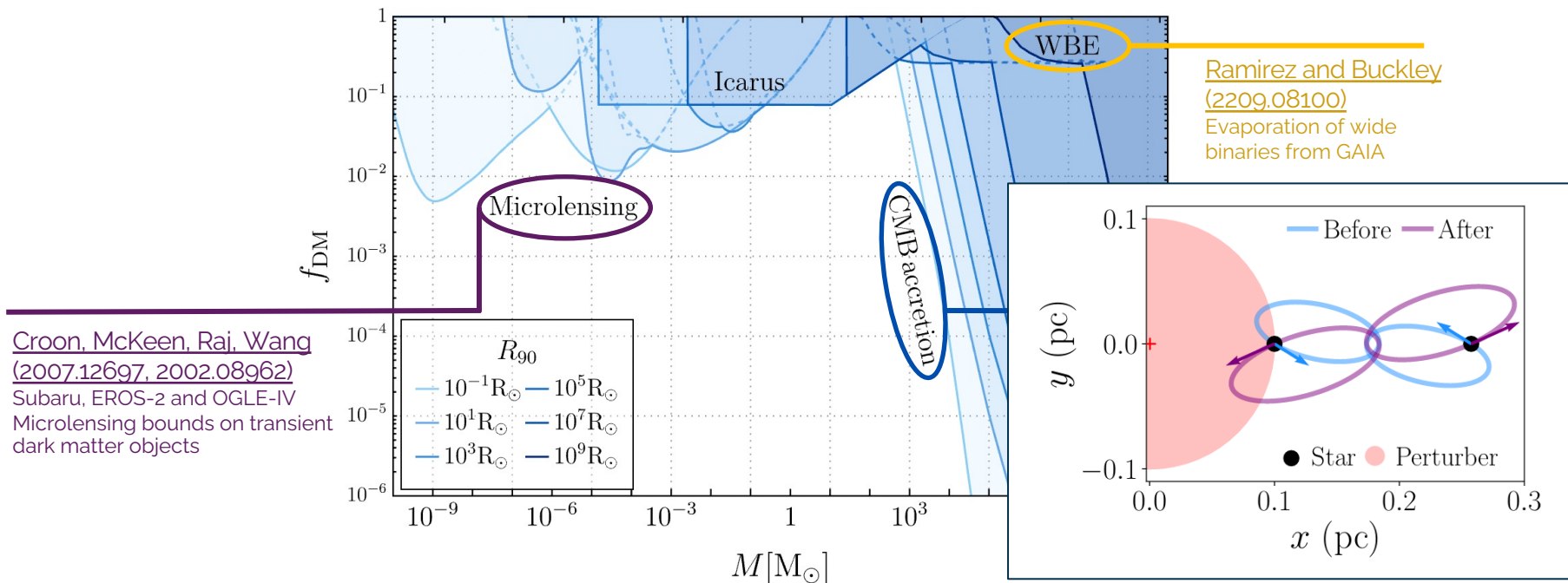
Subaru, EROS-2 and OGLE-IV
Microlensing bounds on transient
dark matter objects

[Croon and SSM \(2403.13072\)](#)
[Bai, Long and Lu \(2003.13182\)](#)

Impact that accretion onto
these object would have on
recombination

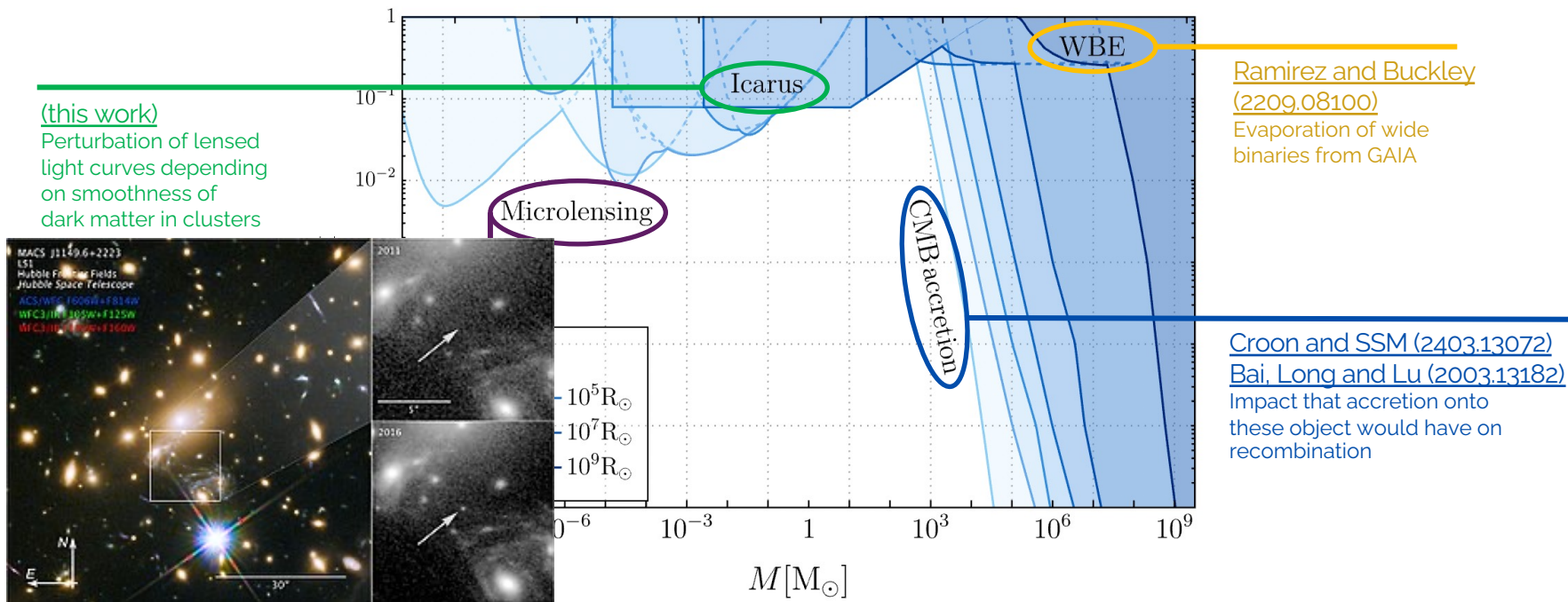
Ultra-compact minihaloes (UCMHs)

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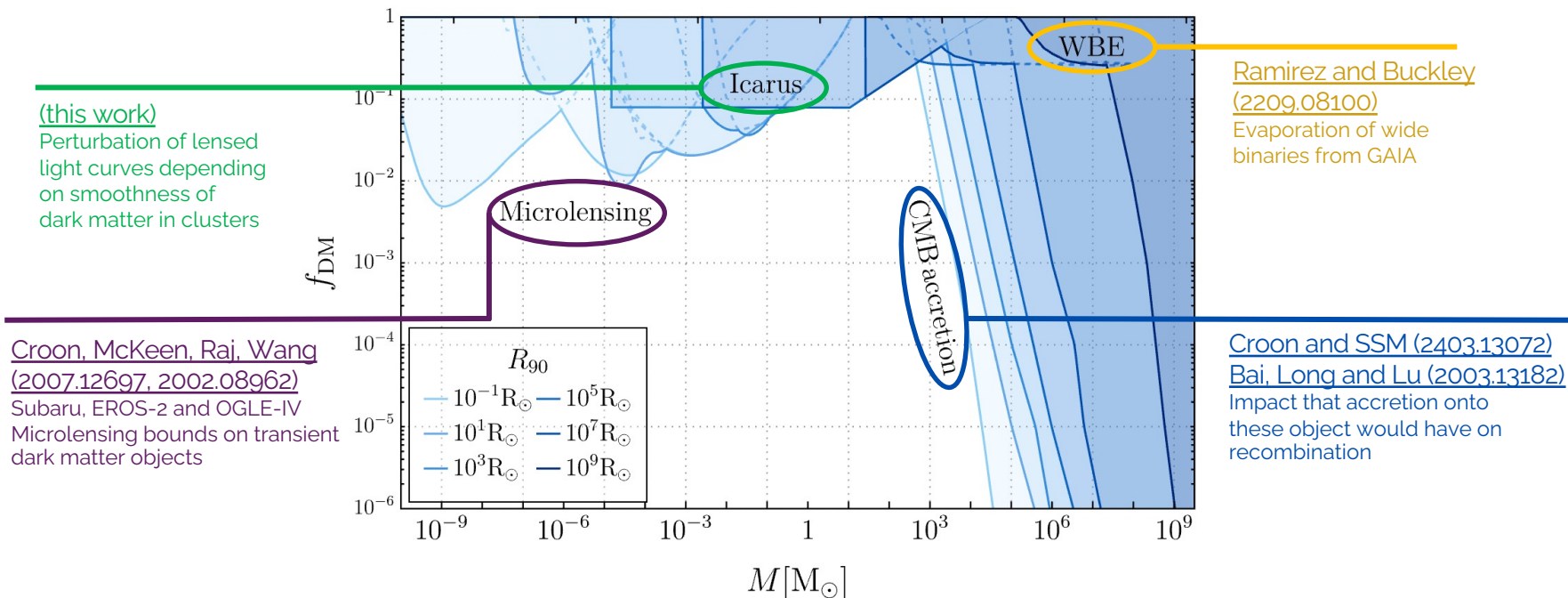
Ultra-compact minihaloes (UCMHs)

There has been recent interest on constraining these type of objects:



Ultra-compact minihaloes (UCMHs)

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Ultra-compact minihaloes (UCMHs)

The object mass, radius and density for a given scale $k \sim 1/R$:

$$M \approx \left[\frac{4\pi}{3} \rho_\chi H^{-3} \right]_{aH=1/R} = \frac{H_0^2}{2G} \Omega_\chi R^3$$

$$\rho_\chi(r, z_c) = \frac{3f_\chi M(z_c)}{16\pi R(z_c)^{3/2} r^{3/2}}$$

$$\frac{R(z_c)}{\text{pc}} = 0.019 \left(\frac{1000}{z_c + 1} \right) \left(\frac{M(z_c)}{M_\odot} \right)^{1/3}$$

where z_c is the collapse redshift

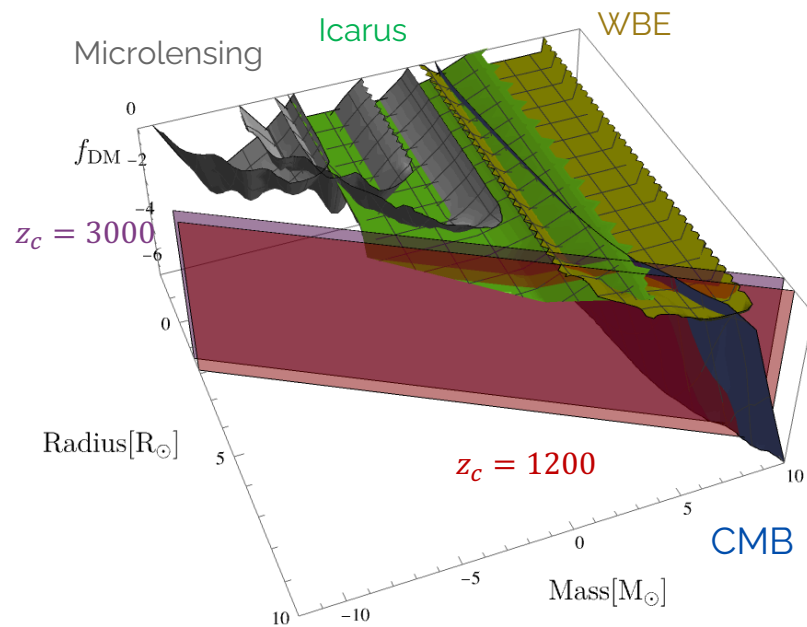
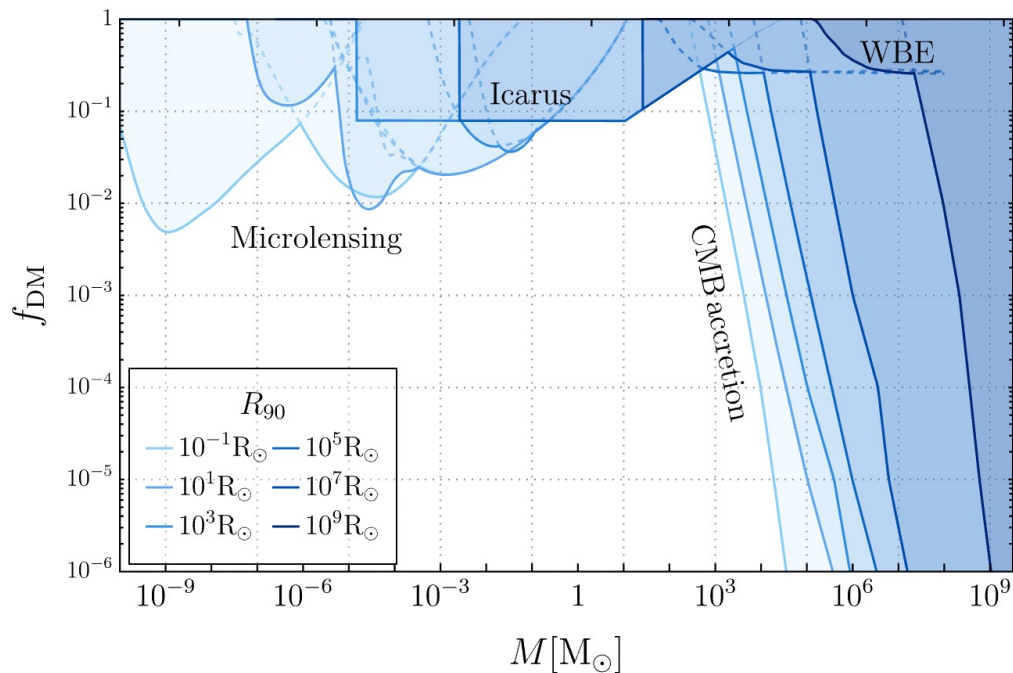


Different density functions have been used, but this is the most conservative

Ultra-compact minihaloes (UCMHs)

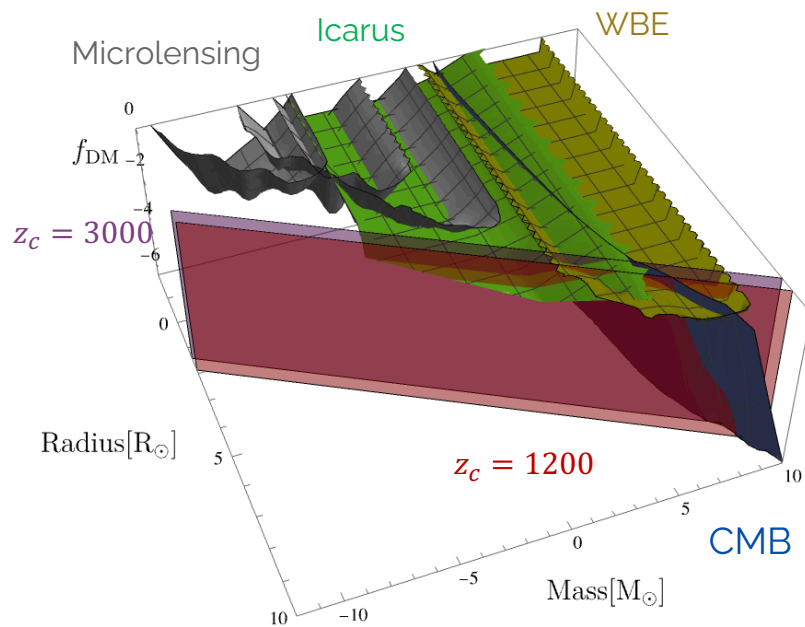
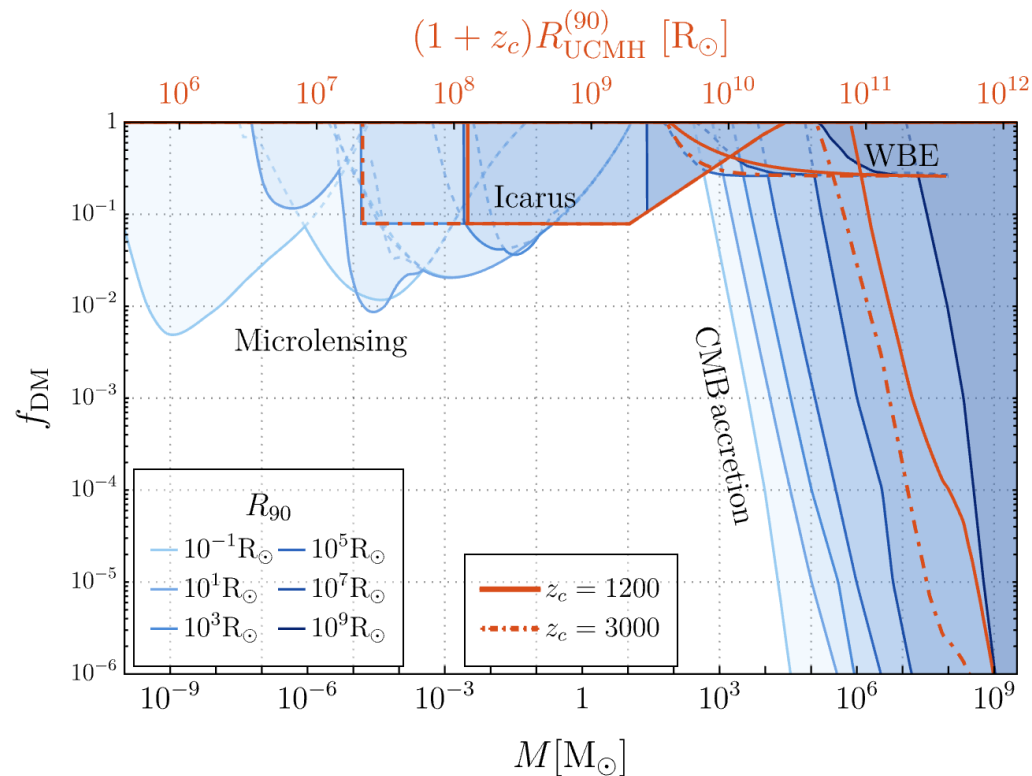
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Only specific objects that have the mass-radius relation can be used to constrain the power spectrum



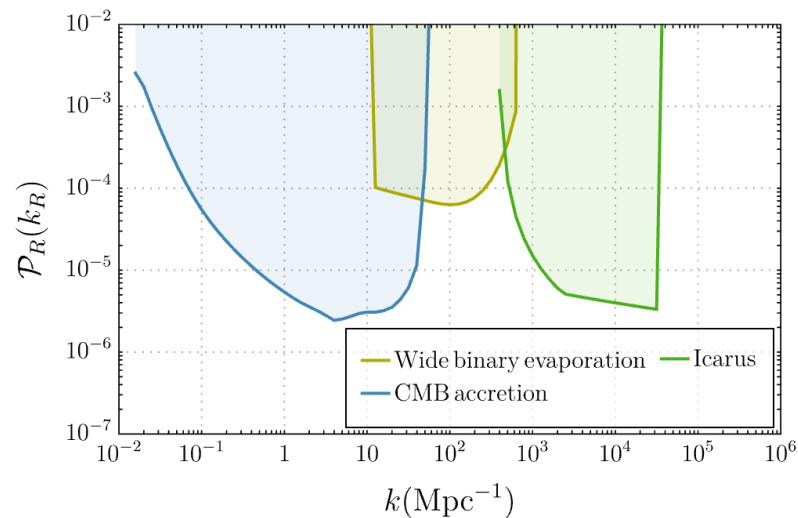
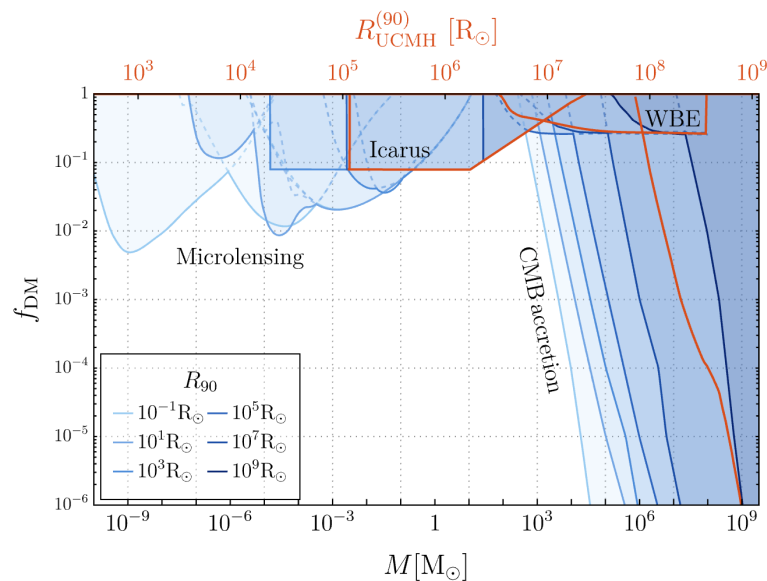
Ultra-compact minihaloes (UCMHs)

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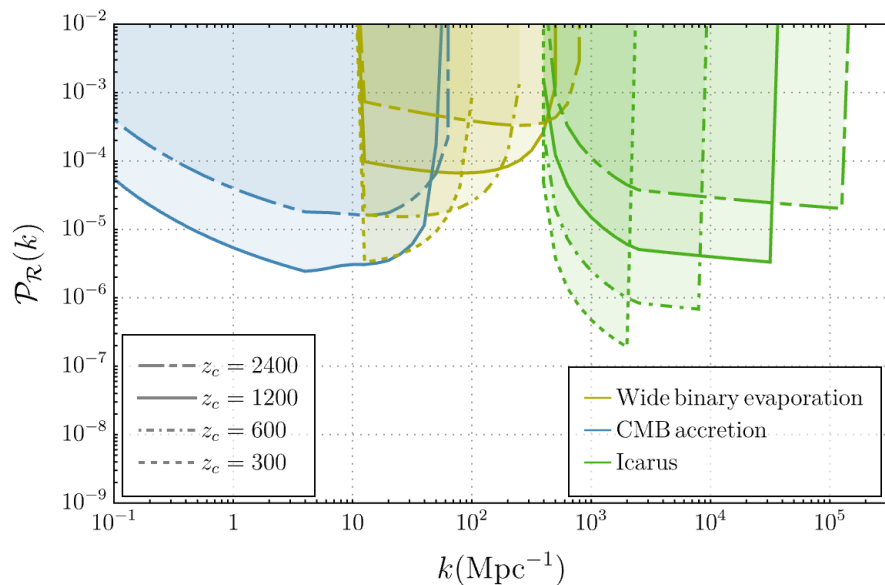
Ultra-compact minihaloes (UCMHs)

We can now translate these bounds (in red) to power spectrum using $\Omega_{\text{UCMH}}(k) = \Omega_{\text{DM}}\beta(k)$



Ultra-compact minihaloes (UCMHs)

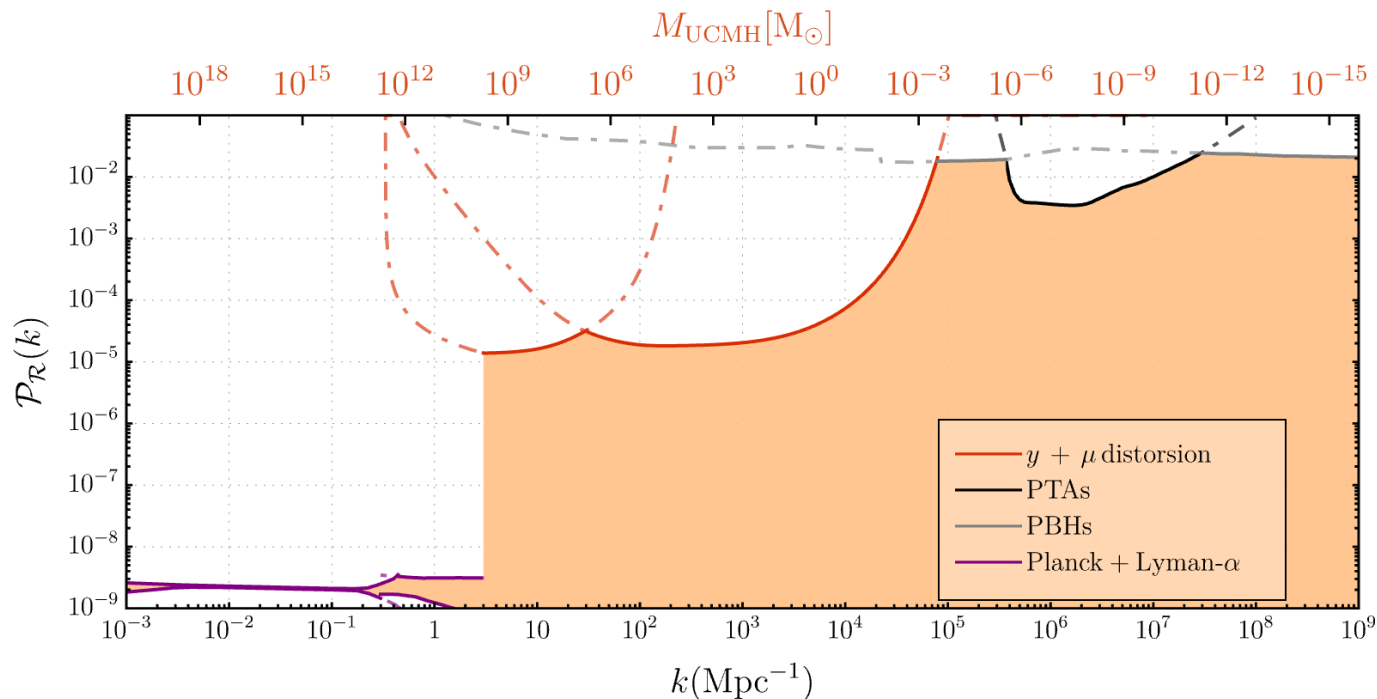
Different collapse times lead to different constraints



We will take the conservative choice $z_c > 1200$

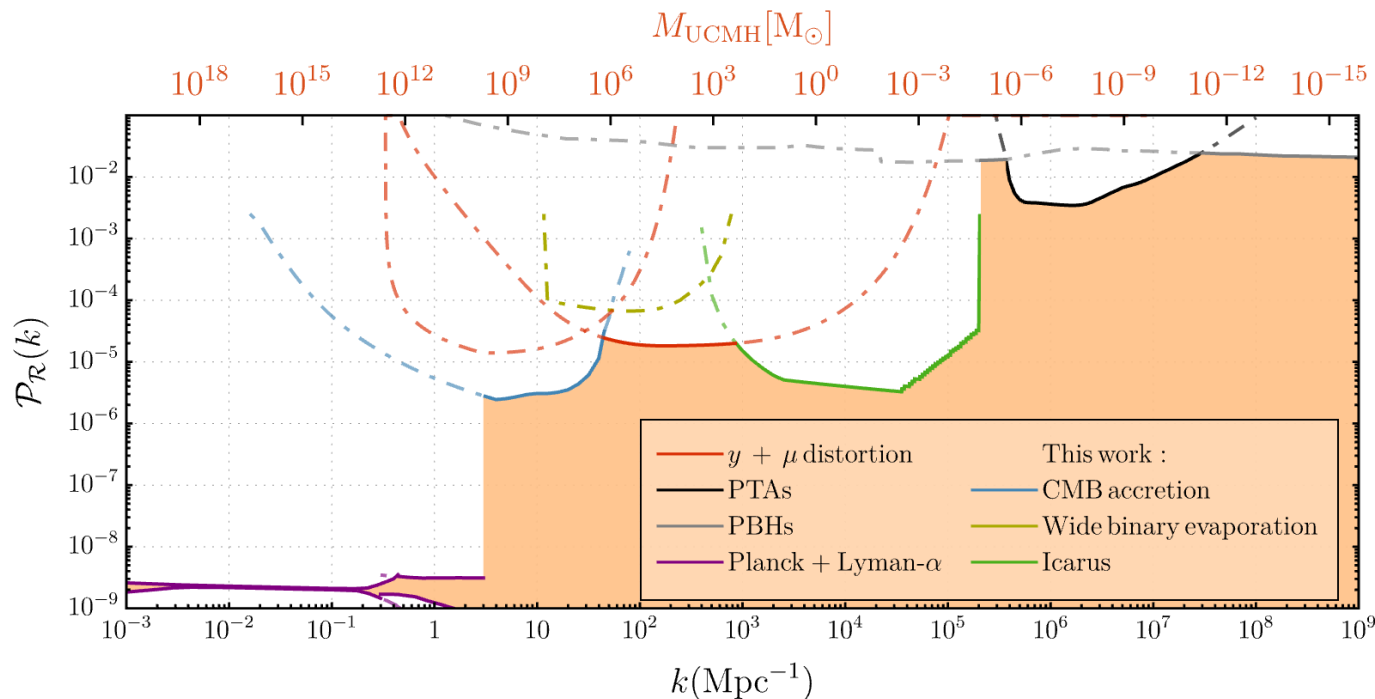
Ultra-compact minihaloes (UCMHs)

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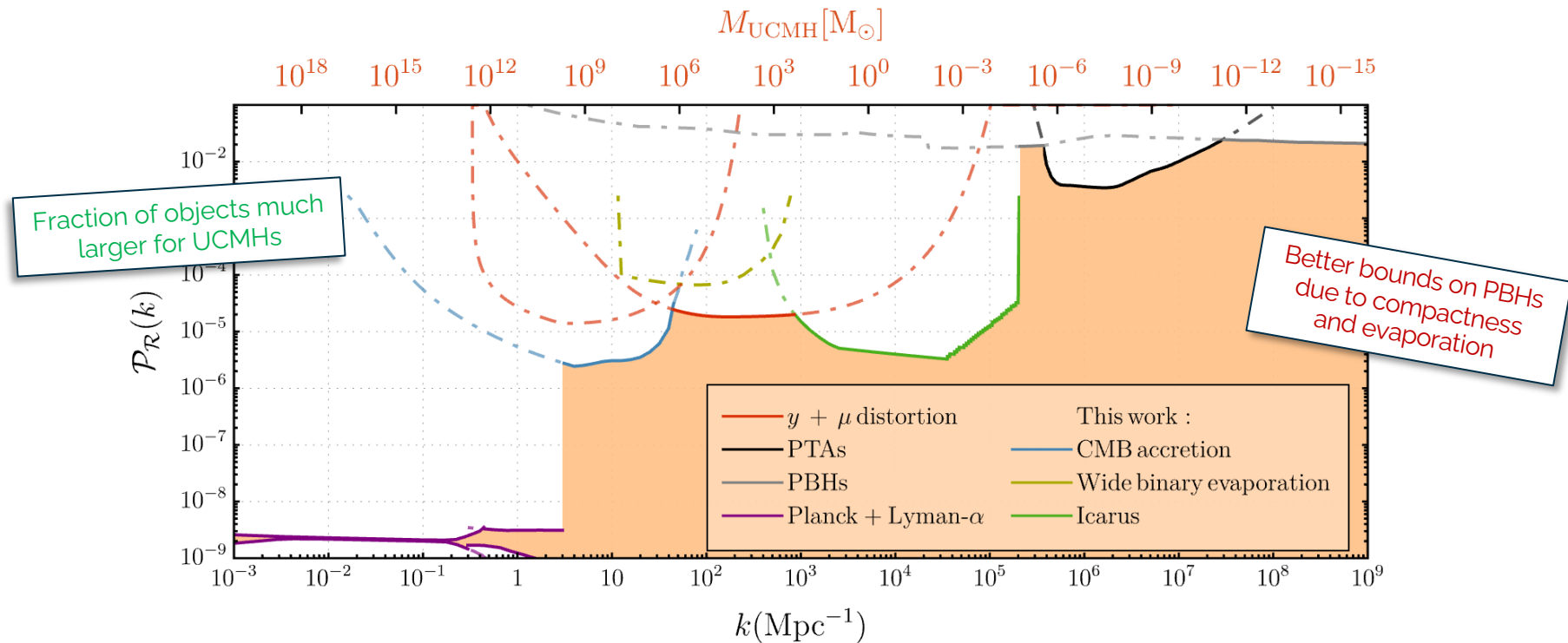


Ultra-compact minihaloes (UCMHs)

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PBHs vs UCMH



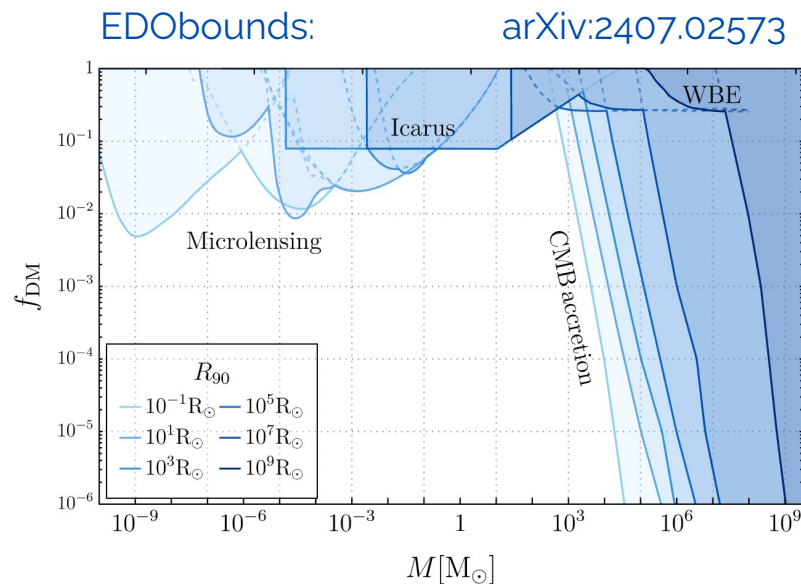
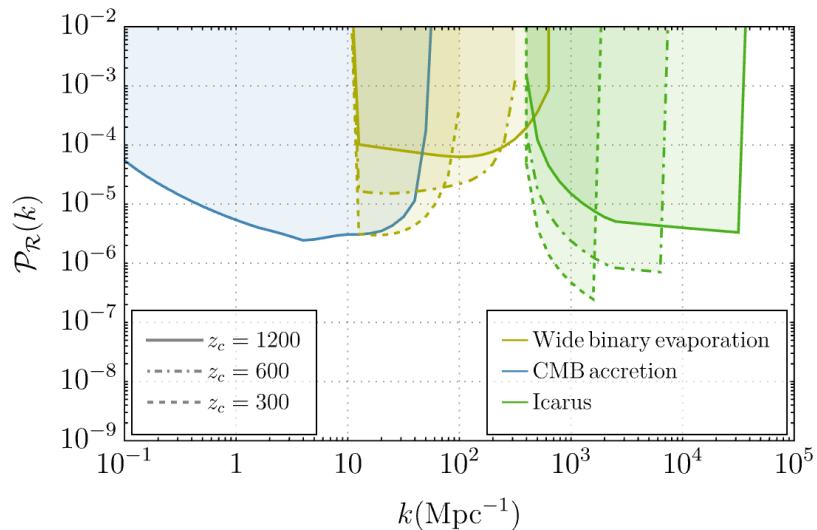
No PBHs with mass $> 100 M_{\odot}$

Soon no masses $> 10^{-1} M_{\odot}$

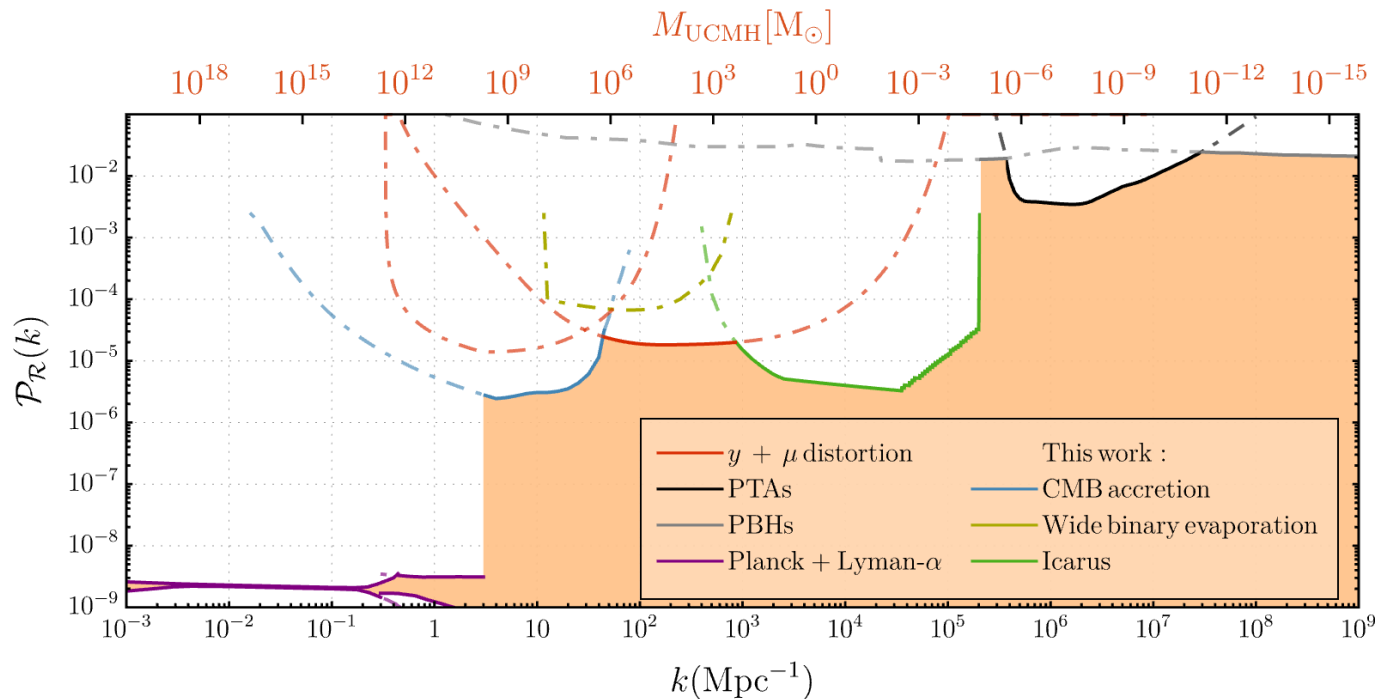
Conclusion

We can obtain improved constraints on the Power Spectrum using UCMHs!

There is room for improvement :)



Thank you for your attention!

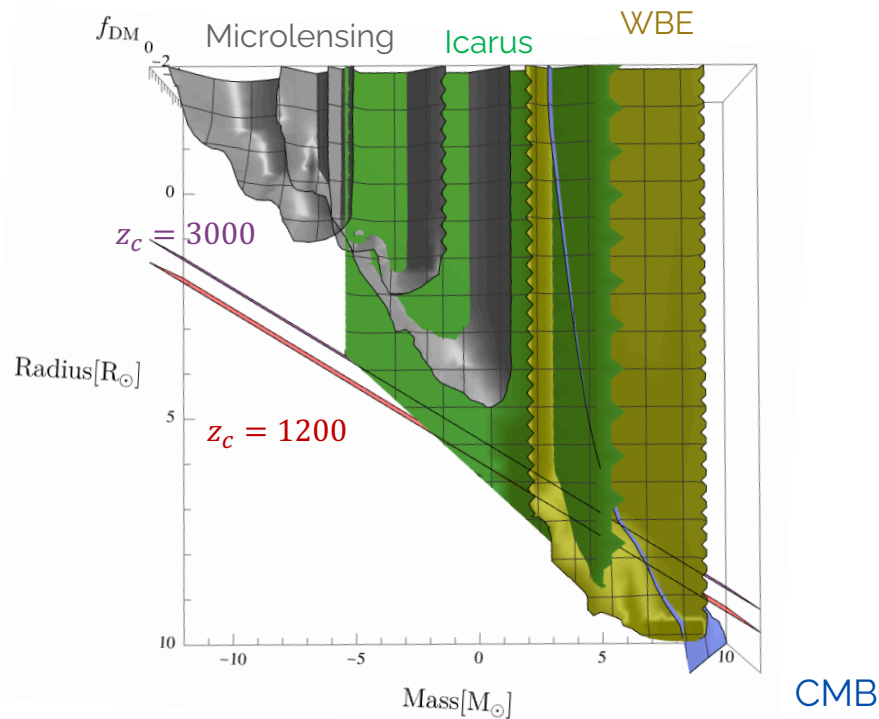
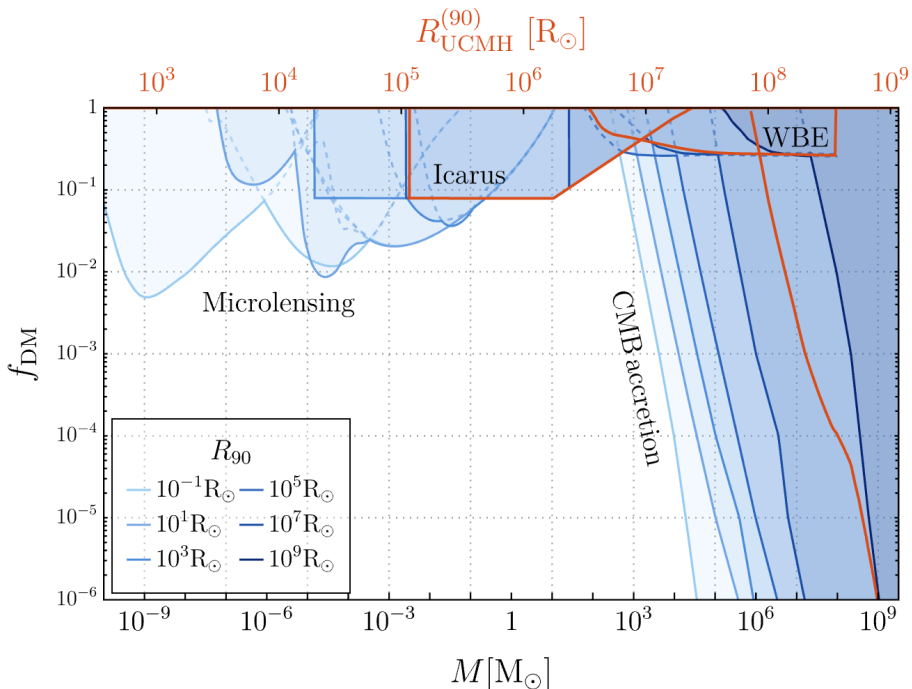


Based on arXiv: 2506.20704

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